GEK-83857

DRIVE SYSTEM INSTRUCTIONS

DRIVE EXTENDER MODULE

DS3820DE1

DRIVE SYSTEMS OPERATIONS
SALEM, VIRGINIA 24153

GENERAL ELECTRIC

DRIVE EXTENDER MODULE

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GENERAL ELECTRIC COMPANY
DRIVE SYSTEMS OPERATIONS
SALEM, VIRGINIA, U.S.A. 24153

ORIGINAL ISSUE DATE: April 1985

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DEFINITION AND FUNCTION

The DS3820DE1 Oil Well Drilling Drive Extender module is a power amplifier unit that provides an increase in brushless field exciter drive for systems that use the DS3820WGC Oil Well Drilling AC Generator Control module (GEM).

THEORY OF OPERATION

SYSTEM OVERVIEW

The System Diagram (Figure 1) outlines the application of the DS3820WGC Generator Engine module, which is used in controlling the power production from a diesel generator set. The system regulates power to the generator's brushless field exciter to control the generator output voltage.

The DS3820DE1 module (DE1) is used in conjunction with the GEM as a power amplifier unit that increases brushless field exciter drive. In its application, the DE1 requires the use of additional components (a field drive transformer, a power supply transformer, an AC snubber, and a DC snubber) in the system.

DE1 MODULE OVERVIEW

The DE1 performs the following system functions: (1) drives a high voltage power transistor by amplifying the GEM's generator voltage regulator output signal (JB), which is normally used to drive the GEM's field transistor: (2) rectifies the field drive transformer output as a power source for the field exciter, which is controlled by the high voltage power transistor; (3) protects against overcurrent; and (4) senses field heat.

The DE1 consists of the following major components (see Figure 2):

- DS3800NRXA Drive Extender board
- Heat Sink Assemblies
  - Transistor heat sink
  - Positive rectifier heat sink
  - Negative rectifier heat sink

NRXA Drive Extender Board

The NRXA board provides the functional circuits described below.

INCREASED DRIVE CIRCUIT. The NRXA's increased drive circuit optically couples and then conditions the JB signal from the GEM module. The conditioned signal is amplified to the required signal level to drive the high voltage power transistor (HVPT).

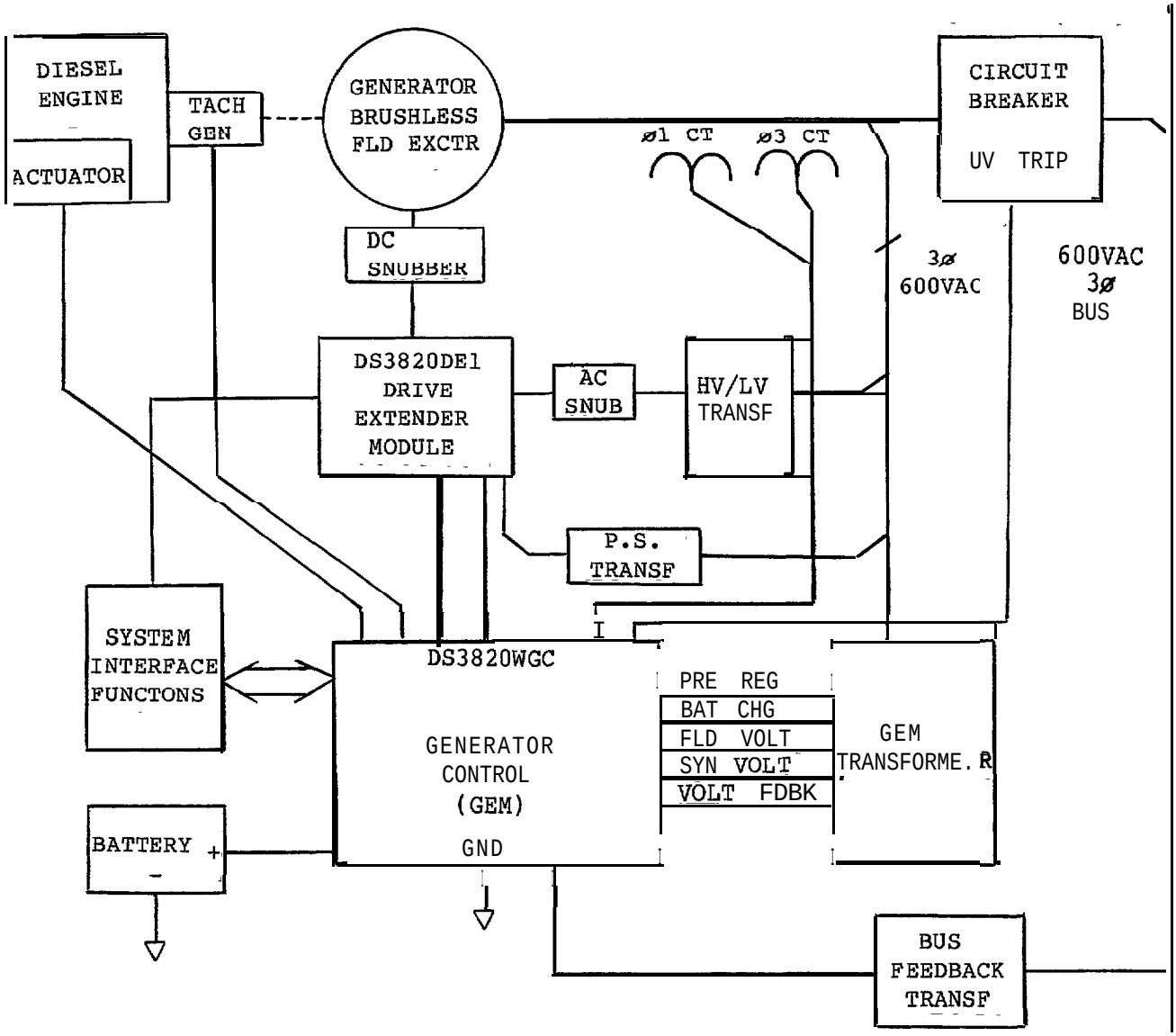


FIGURE 1 SYSTEM DIAGRAM

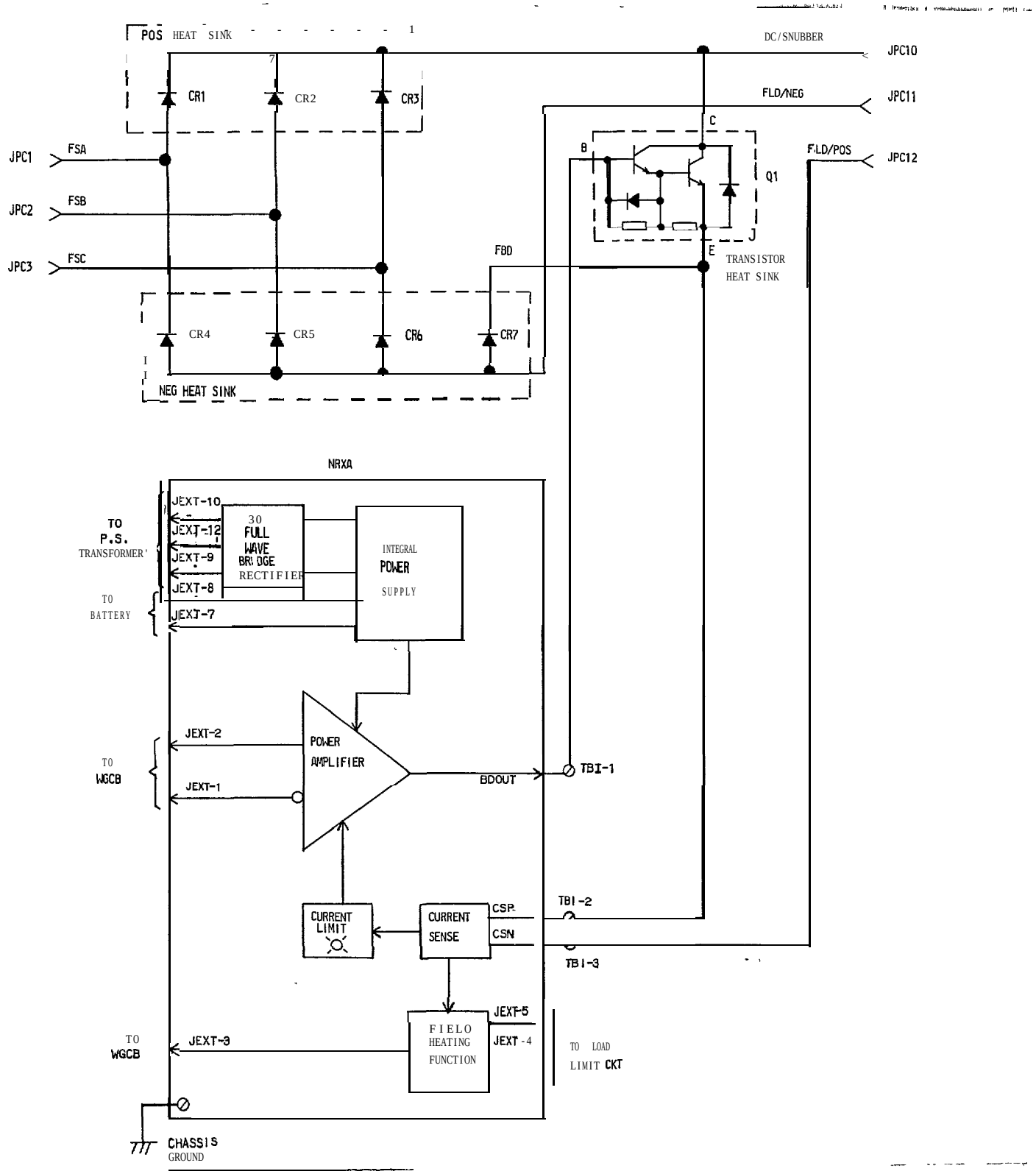


FIGURE 2 DRIVE EXTENDER

CURRENT LIMIT PROTECTION. The NRXA board limits the brushless field exciter load current. The voltage drop across three parallel sensing resistors in series with the HVPT and the load senses the load current. The circuit compares this voltage signal with the selected current limit reference. If the voltage signal equals exceeds the reference, the HVPT drive signal is interrupted to turn off the HVPT. LED 2 indicates when the current limit circuit is active.

FIELD HEATING CIRCUIT. The NRXA's field heating circuit provides a signal proportional to the field temperature. The voltage signal developed from the current limit sensing resistors (and used by the current limit circuit) is also amplified and calibrated by the field heating circuit, such that TP2 equals 5.1v for the maximum average field current. When the field current exceeds 5.1v, the amount of excess is multiplied by 2 and then summed with a voltage input that represents the armature current. This summing produces the required field heating output. (The armature current input is provided by the WGC field heating output.) When the DE1 is powered by battery, the field heating output is disabled and equals approximately zero volts.

POWER SUPPLY. The NRXA contains its own integral power supplies, which are an unregulated +8 volts, +5 volts, and ±12 volts.

The unregulated +8 volt power supply can operate either from the system battery (until the generator voltage is up) or from the rectification of a three-phase power source derived from the generator output. The +5 volt power supply operates from the unregulated +8 volts. The ±12 volt power supply operates from the +5 volt supply.

LED DISPLAY. The NRXA contains the following indicating LEDs:

- LED 1 (POWER SUPPLY OKAY) turns on when the power supply is "okay"
- LED 2 (FIELD CURRENT LIMIT) turns on when Field Current Limit is active
- LED 3 (AC POWER) turns on when the AC power is "on"

DE1 Heat Sink Assembly

The DE1 contains three heat sink assemblies:

- The High Voltage Power Transistor heat sink has only the HVPT mounted on it.
- The Positive Field Rectifier heat sink has three diodes mounted (one per phase voltage).
- The Negative Field Rectifier heat sink has four diodes mounted (three for voltage rectification and one as a flyback diode).

All heat sinks are electrically hot and are mounted on an insulated base located on the DE1.



HEAT SINKS OPERATE AT VERY HIGH VOLTAGE, AND THEREFORE CONTAIN A POTENTIAL HAZARD OF ELECTRICAL SHOCK OR BURN; ONLY PERSONNEL WHO ARE ADEQUATELY TRAINED AND THOROUGHLY FAMILIAR WITH THE EQUIPMENT AND THE INSTRUCTIONS SHOULD INSTALL, OPERATE, OR MAINTAIN THIS EQUIPMENT;

TO PREVENT HAZARD OF ELECTRICAL SHOCK OR BURN, APPROVED GROUNDING PRACTICES AND PROCEDURES MUST BE ADHERED TO;

REQUIRED ADDITIONAL COMPONENTS

Additional components must be used in conjunction with the DE1, as described below.

Brushless Field Exciter Transformers

As a power source for field drive, the DE1 module requries that the system include a low-voltage transformer and a high-voltage transformer. The low-voltage transformer provides a 10 ampere source for driving the brushless field exciter; the high-voltage transformer provides a 5 ampere source for driving the brushless field exciter.

Snubber Circuits

The DE1 requires that the system include an AC snubber circuit and a DC snubber circuit to filter transient voltage spikes on the field transformer. The AC snubber is located on the DE1's input and consists of a capacitor in series with a resistor across each phase-to-phase voltage. The DC snubber circuit consists of a capacitor and a series resistor across the field transformer's rectified secondary output.

SYSTEM APPLICATION

RETROFITS

When an increase in brushless field exciter drive is required by systems that contain the GEM module, the DE1 is used in the system retrofit. The need for additional components depends on the system's requirements, such as brushless field exciter voltage and current limitation.

STANDARD WGC

In addition to retrofitting, the DE1 module can be used when initially constructing systems that apply the GEM module and require additional drive for the brushless field exciter. The need for additional components depends on the system's requirements.

OPERATING AND STORAGE ENVIRONMENT

See GEK-61680.

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SYSTEM OVERVIEW

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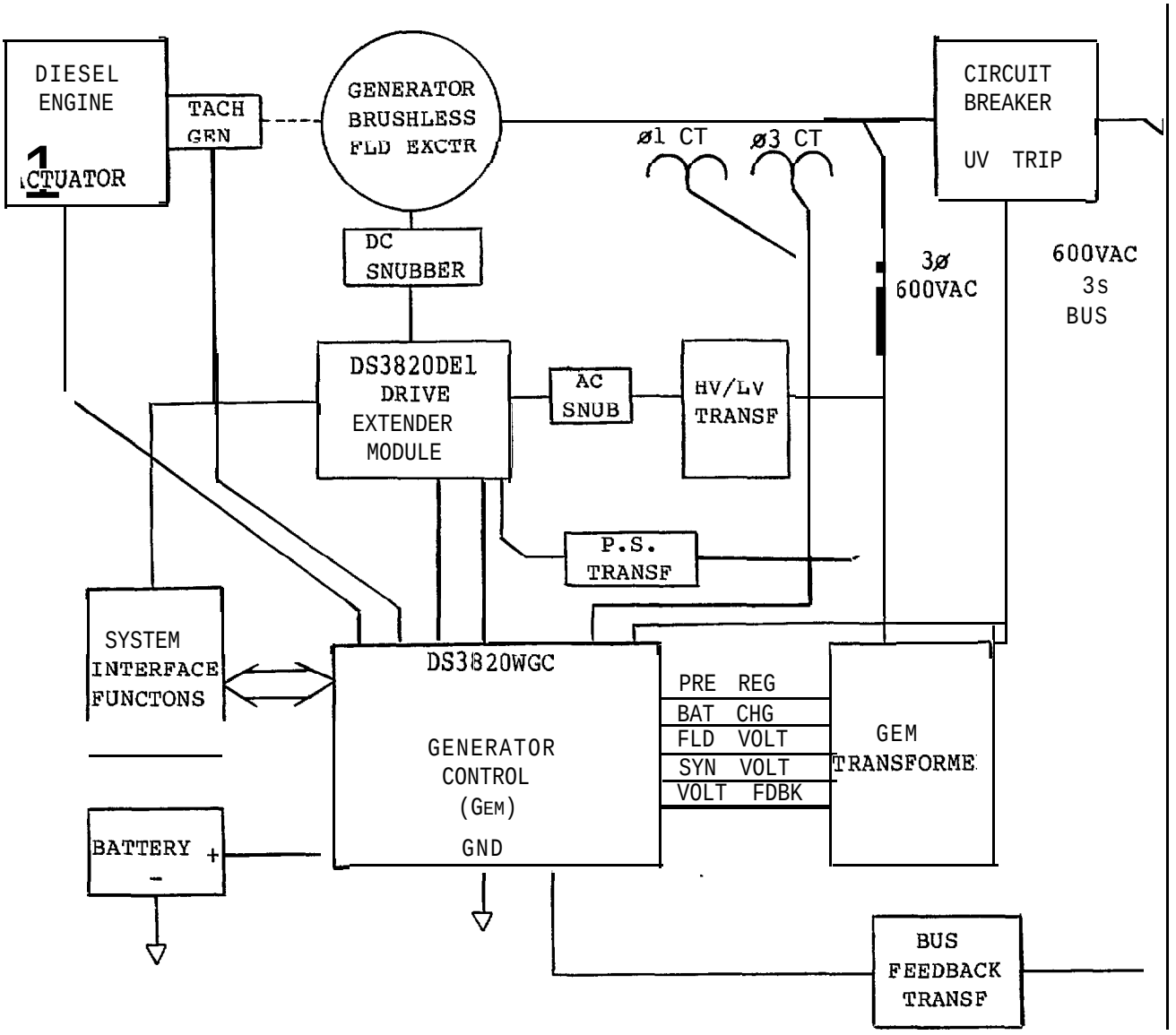


FIGURE 1 SYSTEM DIAGRAM

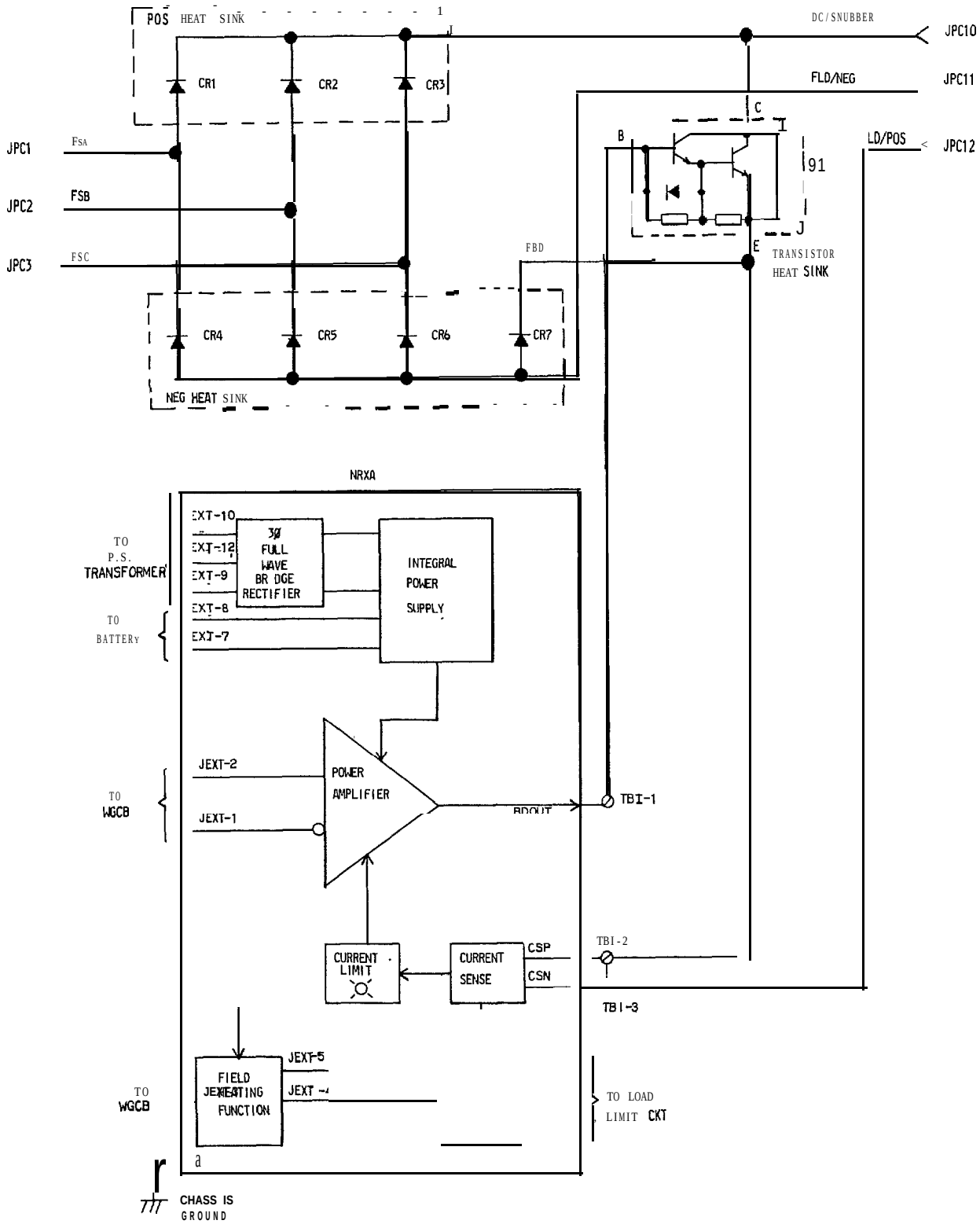


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